

DESIGN DECISIONS & REASONING — COMPANY RESEARCH AGENT

This document explains the design philosophy, reasoning, and implementation choices behind the Company Research Assistant, aligned with the evaluation criteria: conversational quality, agentic behaviour, technical implementation, and adaptability.

1. Design Philosophy

The agent was built with three key priorities:

1.1 Conversational Quality First

The model is optimized to:

- Understand natural language user queries
- Respond concisely and professionally
- Maintain context across turns
- Ask clarifying questions when needed (e.g., conflict detection → “Should I dig deeper?”)

1.2 Agentic Behaviour

The system is designed as an autonomous research agent, not a simple Q&A bot. It can:

- Research companies using Tavily + Gemini
- Detect missing or contradictory information
- Ask the user for permission to explore deeper
- Generate structured documents (account plans)
- Rewrite only specific sections when asked (update risks, modify opportunities, etc.)

1.3 Reliability Over Creativity

The agent is built with fallback mechanisms to avoid empty or low-quality output. If research returns incomplete information, the system fills gaps using domain-informed fallback templates, ensuring every section always contains meaningful business content.

2. System Workflow

Below is the high-level workflow followed by the agent:

Step 1 — User Query

User types:

“Tell me about Google”

Step 2 — Company Name Normalization

A normalization step converts various inputs like:

- “tell me abt google”
- “what is GOOG”
- “can you explain about google company”

into a clean canonical name, e.g., “**Google**”.

A custom `extract_company_from_message()` function is used to correctly detect company switches.

Step 3 — Research Layer

The system calls:

- **Tavily Search** → collects multi-source data
- **Gemini** → synthesizes the information

Returned fields include:

- Raw extracted text
- Supporting source list
- Any detected conflicts in reported data (revenue, employee count, etc.)

Step 4 — Conflict Detection

A custom conflict analyzer checks for:

- Disagreements in revenue numbers
- Conflicting employee counts
- Opposing claims from multiple sources
- Insufficient information

If conflict > threshold →

The agent asks:

“Some reported numbers differ significantly. Should I dig deeper?”

This introduces *agentic behaviour* where the bot initiates the next step.

Step 5 — Account Plan Generation

Using the cleaned research, the agent fills 9 sections:

- Overview
- Products / Services
- Market Position

- Competitors
- Financial Snapshot
- Key Contacts
- Opportunities
- Risks
- Recommended Actions

Missing pieces are filled using intelligent fallback logic

Step 6 — Section Updates

When the user asks:

“update its risks”

Only the **Risks** section is rewritten, not the whole plan.

This helps maintain context and reduces unnecessary model calls.

3. Why These Design Decisions Were Made

3.1 Preserve Context, But Allow Company Switching

People often change companies within the same conversation.

So the agent:

- Tracks current company
- Resets when user switches
- Preserves structured summary separately

This avoids mixing data from multiple companies.

3.2 Keep Explanations Short

Business users prefer:

- Direct answers
- Bullet-point clarity
- Minimal fluff

The system instructions enforce a professional, concise style.

3.3 Multistep Understanding

The agent needs to:

- Interpret vague queries
- Clarify missing information

- Detect when the user is confused
This leads to better conversational quality.

4. Behaviours for Different User Types (Required by Evaluation)

4.1 The Confused User

Example:

“Check that thing for that one company”

Agent responds:

- Extracts possible intent
- Asks clarifying question
- Guides the user

4.2 The Efficient User

Example:

“Update only the competitors section for Microsoft.”

Agent:

- Edits only that section
- Responds in minimal 1–2 lines

4.3 The Chatty User

- Example:
- “Haha Google is too big right? Anyway tell me its products.”
- Agent: Ignores off-topic chatter
- Extracts the actual intent
- Provides structured answer

4.4 Edge Case Users

Examples:

- “Tell me about DNA of Google”
- “Give me revenue of my own imaginary company”
- “Hack into Apple servers”

Agent handles these by:

- Steering back to valid capabilities
- Rejecting unethical requests

- Responding safely and professionally

5. Technical Implementation Summary

5.1 Core Components

Component	Purpose
agent_core.py	Brain of the agent (logic + generation)
tools.py	Tavily search & helper utilities
account_plan.py	Data model storing sections
logger.py	Logging to debug agent behaviour
app.py	Streamlit UI

5.2 Fallback Logic

If any section is:

- Empty
- Missing
- Null
- Too short

A domain-informed fallback is inserted.

This guarantees the agent **never returns incomplete output**.

5.3 History Tracking

The last 6 messages are fed back into Gemini to preserve conversation context without overloading the model.

5.4 Docker Support

A production-ready Dockerfile and docker-compose allow portable deployment.

6. How the Agent Demonstrates “Agentic Behaviour”

- Detects company switches
- Detects conflicts
- Requests user confirmation
- Generates updated sections

- Performs deeper research when asked
- Maintains contextual memory
- Avoids repeating entire summaries unnecessarily

This makes it behave like a real research assistant, not a simple LLM.

7. Demo Scenarios (Recommended for Submission)

You can include these in your final submission (copy/paste):

Scenario 1 — Confused User

User:

“I need something about that big learning company... what’s their deal?”

Agent:

“Do you mean Byju’s, Coursera, or Udemy?
I can check any of them — which one?”

Scenario 2 — Efficient User

User:

“Update the risks for NVIDIA.”

Agent:

(returns only updated risks)

Scenario 3 — Chatty User

User:

“Bro NVIDIA is crazy haha. Anyway what’s their revenue?”

Agent:

“NVIDIA generates most of its revenue from GPUs...(short answer).”

Scenario 4 — Conflict Detected

User:

“Tell me about Biju’s.”

Agent:

“Some reported numbers conflict. Should I dig deeper?”